



ESTIMATIONS OF THE BEST APPROXIMATION OF FUNCTIONS BELONGING TO LIPSCHITZ CLASSES

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Abstract

In this paper, an estimation of the best approximation of functions belonging to Lipschitz classes is provided.

1. Introduction

In this section, we provide some well-known definitions that were previously established in [1-5].

L_p denotes the set of all 2π -periodic functions f , measurable on the segment $[0, 2\pi]$ for $p \in [1, \infty)$ and f is continuous on $[0, 2\pi]$ for $p = \infty$, i.e.,

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$$\|f\|_{L_p} = \begin{cases} \left(\int_0^{2\pi} |f(x)|^p dx \right)^{1/p} < \infty, & \text{for } p \in [1, \infty) \\ \max_x |f(x)|, & \text{for } p = \infty. \end{cases}$$

For $p = \infty$, L_∞ spaces, denoted by C , i.e., the set of continuous functions.

$E_n(f)_p$ denotes the best approximation of the function $f \in L_p$ by trigonometric polynomials with a degree not greater than n , $n \in N$, i.e., $E_n(f)_p = \inf_{H_n} \|f - H_n(x)\|_p$, where $H_n(x)$ are trigonometric polynomials of degree n .

Note 1.1 [4]. From the properties of the best approximation, it is clear that $E_{n+1}(f)_p \leq E_n(f)_p$, $n \in N$.

Theorem 1.1 [4]. Let $f \in CL_2$, $n+1 \in N$, and $S_n(f)$ be the n -th partial sum of the Fourier series of the function f . Then

$$(i) \ E_n(f)_2 = \|f - S_n(f)\|_2,$$

$$(ii) \ \text{If for } T \in H_n, \|f - T\|_2 = E_n(f)_2, \text{ then } T = S_n(f),$$

$$(iii) \ E_n(f)_2 = \left(\pi \sum_{m=n+1}^{\infty} (a_m^2 + b_m^2) \right)^{\frac{1}{2}}.$$

Definition 1.1 [4]. Let $f \in L_1$, $m, n+1 \in N$. The Vallee-Poussin partial sum of the function f , is said to be the sum

$$\sigma_{n,m}(f) = \frac{\sum_{k=n}^{n+m-1} S_k(f)}{m},$$

where $S_k(f)$ represents k -th partial sum of the Fourier series.

Note 1.2. The following statements are true for the Vallee-Poussin partial sum of the function f :

$$(i) \sigma_{n,1}(f) = S_n(f), \sigma_{0,m+1}(f) = \sigma_m(f);$$

$$(ii) \sigma_{n,m}(f) = \left(1 + \frac{n}{m}\right) \sigma_{n+m-1}(f) - \frac{n}{m} \sigma_{n-1}(f);$$

$$(iii) \|\sigma_n(f)\|_1 \leq \|f\|_1;$$

$$(iv) \|f(x) - \sigma_{n,m}(f(x))\|_C \leq \left(\frac{4}{\pi^2} \ln\left(1 + \frac{n}{m}\right) + 3\right) E_n(f)_C.$$

Let $\alpha, M > 0$ and $f \in C$. We say that the function $f \in C$ satisfies a Lipschitz condition of order α related to the constant M in spaces C if $\omega(f, t) \leq M|t|^\alpha$ and symbolically is written that $f \in Lip_M(\alpha, C)$ or simply $f \in Lip \alpha$.

2. Main Results

Theorem 2.1. Let $\frac{a_0}{2} + \sum_{i=1}^{\infty} a_i \cos ix + b_i \sin ix$ be the Fourier series of the function $f \in Lip_M(\alpha, C)$. Then $E_n(f)_2 \leq Mn^{-\alpha}$.

Proof. In [6], we have obtained that

$$\sum_{i=n}^{2n-1} a_i^2 + b_i^2 \leq M_1 n^{-2\alpha},$$

where M_1 is a positive constant. Using the properties of $E_n(f)_2$ and changing the summation order, we get

$$\frac{E_n(f)_2}{\sqrt{\pi}} \leq \left\{ \sum_{t=0}^{\infty} \sum_{i=2^t n}^{2^{t+1} n-1} (a_i^2 + b_i^2) \right\}^{1/2} \leq \left\{ \sum_{t=0}^{\infty} \frac{M_1 n^{-2\alpha}}{2^{2\alpha t}} \right\}^{1/2}$$

$$= M_1 n^{-\alpha} \left\{ \sum_{t=0}^{\infty} \frac{1}{4^{\alpha t}} \right\}^{1/2} \leq M n^{-\alpha}.$$

$$\Rightarrow E_n(f)_2 \leq \sqrt{\pi} M_1 n^{-\alpha} = M n^{-\alpha}.$$

Theorem 2.2. *If the function $f \in C$ satisfies the relation*

$$\sum_{k=n}^{\infty} \left(\frac{E_n(f)_2}{\sqrt{n}} \right) \leq M n^{-\alpha}, \text{ where } M \text{ is a positive constant and } 0 \leq \alpha < 1, \text{ then}$$

$f \in Lip \alpha$.

Proof. Using the Holder inequality, the relation

$$\sum_{i=n}^{2n-1} a_i^2 + b_i^2 \leq M_1 n^{-2\alpha},$$

and the relation for the best approximation (see [7]), we get the following relation:

$$\begin{aligned} & |f(t+h) - f(t-h)| \\ &= \left| 2 \sum_{i=1}^{\infty} (b_i \cos ix - a_i \sin ix) \sin ih \right| \\ &\leq 2 \sum_{t=1}^n \sum_{i=2^{t-1}}^{2^t-1} (|a_i| + |b_i|) |h| + 2 \sum_{i=2^n}^{\infty} (|a_i| + |b_i|) \\ &\leq 2|h| \sum_{t=1}^n \left(\sum_{i=2^{t-1}}^{2^t-1} (|a_i|^2 + |b_i|^2) \right)^{1/2} \left(\sum_{i=2^{t-1}}^{2^t-1} (1^2) \right)^{1/2} \\ &\quad + 2\sqrt{2} \sum_{i=2^n}^{\infty} \left(\frac{|a_i|^2 + |b_i|^2}{2} \right)^{1/2} \end{aligned}$$

$$\begin{aligned} &\leq M_1 |h| \sum_{t=1}^n ((2^{-2\alpha(t-1)})^{1/2} ((2^{t-1})^2)^{1-1/2}) + \frac{2}{\sqrt{\pi}} \sum_{m=2^n}^{\infty} \left(\frac{E_m(f)_2}{\sqrt{m}} \right) \\ &\leq M_1 |h| \sum_{t=1}^n (2^{-\alpha(t-1)} 2^{t-1}) + \frac{2}{\sqrt{\pi}} \frac{1}{2^{\alpha n}} \leq M_2 \left(\frac{|h|}{2^{(\alpha-1)n}} + \frac{1}{2^{\alpha n}} \right). \end{aligned}$$

Now, choosing an integer n , such that $|h| \in \left(\frac{1}{2^n}, \frac{1}{2^{n-1}} \right]$, we get the relation

$$|f(t+h) - f(t-h)| \leq \frac{M_3}{2^{\alpha n}} \leq M_3 |h|^\alpha,$$

where M_i , $i = 1, 2, 3$ represent positive constants, i.e., $f \in Lip \alpha$.

Theorem 2.3. *If the function $f \in C$ satisfies the relation*

$$\sum_{k=n}^{\infty} \sqrt{\frac{E_n(f)_C \|f(x)\|_1}{n}} \leq M n^{-\alpha},$$

where M is a positive constant and $0 \leq \alpha < 1$, then $f \in Lip \alpha$.

Proof.

$$\begin{aligned} &|f(t+h) - f(t-h)| \\ &= \left| 2 \sum_{i=1}^{\infty} (b_i \cos ix - a_i \sin ix) \sin ih \right| \\ &\leq 2 \sum_{t=1}^n \sum_{i=2^{t-1}}^{2^t-1} (|a_i| + |b_i|) |h| + 2 \sum_{i=2^n}^{\infty} (|a_i| + |b_i|) \\ &\leq M_1 \frac{|h|}{2^{(\alpha-1)n}} + \frac{2}{\sqrt{\pi}} \sum_{m=2^n}^{\infty} \left(\frac{E_{2m-1}(f)_2 + E_{2m}(f)_2}{\sqrt{m}} \right) \end{aligned}$$

$$\begin{aligned}
&\leq M_1 \frac{|h|}{2^{(\alpha-1)n}} + \frac{4}{\sqrt{\pi}} \sum_{m=2^n}^{\infty} \left(\frac{E_{2m-1}(f)_2}{\sqrt{m}} \right) \\
&= M_1 \frac{|h|}{2^{(\alpha-1)n}} + \frac{4}{\sqrt{\pi}} \sum_{m=2^n}^{\infty} \left(\frac{\left(\int_{-\pi}^{\pi} (f(x) - \sigma_{m,m}(f(x)))^2 dx \right)^{\frac{1}{2}}}{\sqrt{m}} \right) \\
&\leq M_1 \frac{|h|}{2^{(\alpha-1)n}} + \frac{4}{\sqrt{\pi}} \\
&\quad \times \sum_{m=2^n}^{\infty} \left(\frac{\max_{x \in R} (|f(x) - \sigma_{m,m}(f(x))|)^{\frac{1}{2}} \left(\int_{-\pi}^{\pi} (f(x) - \sigma_{m,m}(f(x))) dx \right)^{\frac{1}{2}}}{\sqrt{m}} \right) \\
&= M_1 \frac{|h|}{2^{(\alpha-1)n}} + \frac{4}{\sqrt{\pi}} \\
&\quad \times \sum_{m=2^n}^{\infty} \left(\frac{(\|f(x) - \sigma_{m,m}(f(x))\|_C \|f(x) - \sigma_{m,m}(f(x))\|_1)^{\frac{1}{2}}}{\sqrt{m}} \right) \\
&\leq M_1 \frac{|h|}{2^{(\alpha-1)n}} + \frac{4}{\sqrt{\pi}} \\
&\quad \left(\left(\frac{4}{\pi^2} \ln \left(1 + \frac{m}{m} \right) + 3 \right) E_m(f)_C \right. \\
&\quad \left. \times \sum_{m=2^n}^{\infty} \frac{\left(\|f(x)\|_1 + 2\|\sigma_{2m-1}(f(x))\|_1 + \|\sigma_{2m-1}(f(x))\|_1 \right)^{\frac{1}{2}}}{\sqrt{m}} \right)
\end{aligned}$$

$$\begin{aligned}
 &\leq M_1 \frac{|h|}{2^{(\alpha-1)n}} + \frac{8}{\sqrt{\pi}} \sqrt{\frac{4}{\pi^2} \ln 2 + 3} \sum_{m=2^n}^{\infty} \frac{(E_m(f)_C \|f(x)\|_1)^{\frac{1}{2}}}{\sqrt{m}} \\
 &\leq M_1 \frac{|h|}{2^{(\alpha-1)n}} + \frac{8}{\sqrt{\pi}} \sqrt{\frac{4}{\pi^2} \ln 2 + 3} \frac{1}{2^{\alpha n}} \\
 &\leq M_2 \left(\frac{|h|}{2^{(\alpha-1)n}} + \frac{1}{2^{\alpha n}} \right).
 \end{aligned}$$

Now, choosing an integer n , such that $|h| \in \left(\frac{1}{2^n}, \frac{1}{2^{n-1}} \right]$, we get the relation

$$|f(t+h) - f(t-h)| \leq \frac{M_3}{2^{\alpha n}} \leq M_3 |h|^\alpha,$$

where M_i , $i = 1, 2, 3$ represent positive constants, i.e., $f \in Lip \alpha$.

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